

APURBA DEBNATH

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EDUCATION

University of California, San Francisco

San Francisco, CA

Master of Science in Biomedical Imaging | GPA: 3.9/4.0

SEP 2025 – SEP 2026

Thesis: *Spectral Graph modeling of abnormal neural synchronizations in Alzheimer's disease.*

National Institute of Technology Durgapur

Durgapur, India

Bachelor of Technology in Biotechnology

DEC 2020 – MAY 2024

Major GPA: 3.78/4.00, CS GPA: 4.0/4.0.

Thesis: *Machine learning-based approach for the design and release kinetics prediction of the PLGA-based drug delivery systems.*

PUBLICATIONS

1. **Debnath, A.,** Venot, T., Corsi, M-C. & Verma, P. (2025). “*Neural mechanisms of training in Brain-Computer Interface: A Biophysical modeling approach,*”(under review) (<https://doi.org/10.1101/2025.06.21.660834>)
2. **Debnath, A.** (2026). “*Spectral Graph modeling of abnormal neural synchronizations in Alzheimer's disease.*” (MS Thesis, UCSF) [PDF](#)
3. **Debnath, A.*,** Barron, N.*, Prabhu, P.,...,Nagarajan, S., & Raj, A., “*Altered frequency-dependent neural driving signals significantly impair E/I dynamics in Alzheimer's disease.*” (in prep.)

RESEARCH EXPERIENCE

University of California, San Francisco, San Francisco, CA, USA

JAN 2026 – SEP 2026

Graduate Student Researcher, Brain Networks Laboratory, UCSF

Advisors: Prof. Ashish Raj & Prof. Srikantan Nagarajan

- Designed and built signal-processing and spectral-analysis pipelines for large-scale EEG/MEG recordings from healthy individuals and patients with Alzheimer's disease (AD). Implemented alternating two-step iterative optimization algorithm based on spectral graph model to deconvolve scale-free neural driving signals and model oscillatory brain activity, and achieved > 0.90 Pearson correlation and < 0.5 RMSE between modeled and empirical spectra for most brain region.
- Quantified deconvolved scale-free neural driving signals and demonstrated that AD pathology (such as FDG-uptake, $A\beta$ and tau accumulation) was accompanied by significant alteration in scale-free neural driving dynamics, with spatial alterations of scale-free signals in the delta-theta (1-7 Hz), alpha (8-12 Hz) and beta (13-30 Hz) band recapitulating regions selectively vulnerable to AD pathology. Moreover, these altered scale-free neural driving signals were also associated with disrupted excitation-inhibition (E/I) dynamics, with AD exhibiting significantly increased excitatory population time constants, where a greater cognitive decline showed network level temporal dysfunction with significantly increased long-range excitatory population time constants.

– **Research Area:** Alzheimer's disease, Computational Neuroscience.

Indian Institute of Technology Madras, Chennai, India

JUL 2024 – JUL 2025

Post-Baccalaureate Researcher, Centre for Integrative Biology and Systems Medicine (IBSE)

Co-Advisors: Dr. Parul Verma (IIT Madras) & Dr. Marie-Constance Corsi (Paris Brain Institute)

- Processed and analyzed longitudinal EEG/MEG recordings from a cohort of 19 healthy participants across four motor imagery (MI)-based BCI training sessions, along with an independent EEG MI-BCI dataset of 16 participants trained under a separate BCI paradigm. Developed a biophysical neural mass model with explicit modeling of scale-free/aperiodic brain activity to investigate the neural mechanisms underlying MI-BCI learning, and achieved > 0.90 Pearson correlation and < 0.3 RMSE between modeled and empirical M/EEG power spectra across most brain regions, subjects, and sessions in both MI and resting-state (RS) conditions.

- Demonstrated that MI condition was accompanied by significant alteration in scale-free brain activity compared to RS, and this task-dependent significant modulation revealed training induced regional E/I reconfiguration, where BCI training drove a shift from diffuse cortical activation to targeted sensorimotor recruitment via regional E/I dynamics. Moreover, the regional activation was accompanied by significant reduction in excitatory population time constant during MI than RS, and MI-related desynchronization/synchronization phenomena was also accompanied by a significant association with excitatory population time.

– **Research Area:** *Brain-Computer Interface, Computational Neuroscience.*

National Institute of Technology Durgapur, India

AUG 2023 – MAY 2024

Undergraduate Researcher, Department of Biotechnology

Advisor: Prof. Dalia Dasgupta Mandal

- Reviewed and analyzed mathematical models to characterize and uncover mechanisms underlying kinetics of drug release, and implemented 8 ML regression models to predict drug release kinetics (fractional drug release) in PLGA-based drug delivery systems (DDSs) and identify key factors affecting the kinetics of drug release, using ≈ 2000 experimental historical drug release profiles having 20 drug–polymer formulations of anticancer and non-anticancer drugs. Designed end-to-end predictive modeling pipelines, where Random Forest and XGBoost demonstrated strong predictive performance for early-stage drug release kinetics (up to $\approx 50\%$ of cumulative drug release).
- Optimized Random Forest model achieved a test R^2 of 0.71, MSE of 0.0264, and MAE of 0.1169, while reducing the feature space from 17 to 7 key physicochemical features, and identified molecular weight, drug loading capacity, surface-area-to-volume ratio, and time as the key determinants of kinetics of drug release in PLGA-based DDS.

– **Research Area:** *Drug Delivery Systems, Machine Learning*

SELECT PROJECTS

Neural time-series decoding using deep learning

UCSF | 2026 – Present

- Develop preprocessing pipelines for multi-modal electrophysiological time-series data and train various deep-learning models on time-series data to decode the neural states and classify.

Automated 3D MRI volumetric segmentation of Gliomas

UCSF | 2026 – Present

- Developed preprocessing pipelines for multimodal 3D MRI (T1, T2, T2-FLAIR, and T2-GD) NIfTI images, including spatial normalization, foreground cropping, patch-based training preparation etc.
- Trained and evaluated 3D UNet and SegResNet models for glioma sub regions, multi-class segmentation using data from 97 subjects, achieving a cross-validation Dice score of 0.95 and test-set Dice scores of approximately $\in [0.80-0.90]$ across subjects and for individual subjects.

Decoding Cognitive tasks from functional-MRI using machine learning

UCSF | 2026

- Designed preprocessing pipelines to preprocess/process multi-session task-based (language and motor task) fMRI recordings from 10 healthy subjects using fMRIPrep.
- Trained a Support Vector Machine (SVM) classifier on the voxel-level preprocessed BOLD signals to decode and classify different language and motor tasks across 10 healthy subjects. Trained SVM achieved an overall accuracy of 86.1% across tasks and subjects, with a strong performance in decoding language tasks, achieving an F1-score of 0.95, and moderate performance in classifying motor tasks, with an F1-score of 0.77. Notably, 21% of motor task instances were misclassified as language tasks.

Deep learning models for brain MRI denoising: A comparative benchmark study

UCSF | 2025 – 2026

- Designed and implemented deep learning modeling pipelines to train and benchmark denoising autoencoders (CAE, VAE, ResNet, and UNet) using 7000 T1 MRI slices from 128 subjects, under both fixed and variable noise conditions.
- ResNet and UNet consistently achieved strong performance, with SSIM values reaching $\sim 0.92-0.96$, PSNR $\sim 30-34$ dB, and EPI $\sim 0.85-0.92$ across noise levels. In contrast, CAE demonstrated moderate performance (SSIM $\sim 0.85-0.91$, PSNR $\sim 27-31$ dB, EPI $\sim 0.78-0.88$), while VAE showed the lowest scores (SSIM $\sim 0.75-0.88$, PSNR $\sim 25-29$ dB, EPI $\sim 0.70-0.82$) with higher variability.

SKILLS

Technical Skills

- **Programming:** Python, MATLAB, C/C++, SQL, $\text{\LaTeX}$
- **Python Packages:** NumPy, Pandas, SciPy, Statsmodels, Pingouin, Matplotlib, MNE, Nilearn
- **Neuroimaging & Neuroscience:** EEG, MEG, MRI (sMRI, fMRI, dMRI), Signal & Image Processing, Signal Decoding
- **Neuroimaging Software:** fMRIPrep, FSL/FAST, MRtrix3, XCP-D, 3D Slicer
- **Machine Learning & Deep Learning:** PyTorch, TensorFlow, Scikit-learn
- **Computing & Tools:** Linux, Git/GitHub, High-Performance Computing (CoreHPC)

Wet Lab

- **Laboratory:** Cell Culture, PCR, Genomic/Plasmid DNA Isolation, Restriction Digestion, Agarose Gel Electrophoresis, ELISA, Spectrophotometry, Centrifugation, Microscopy, Growth Kinetics

SELECTED CONFERENCE POSTERS

1. **Debnath, A.,** Barron, N., Raj, A. (2026). **Spectral Graph modeling of abnormal neural synchronizations in Alzheimer's disease.** *2026 UCSF Annual Radiology Research Conference, Santa Cruz, CA. Poster: March 30, 2026.*
2. **Debnath, A.,** Verma, P., Corsi, M-C. (2025). **A biophysical model to infer neural mechanisms of motor imagery in brain-computer interface training.** *OHBM 2025 Annual Meeting, Brisbane, Australia. Poster: June 27-28, 2025.*
3. **Debnath, A.,** Corsi, M-C., Verma, P. (2025). **A Neural mass modeling approach for uncovering neural mechanisms in motor imagery-based brain-computer interface.** *Wadhvani School of Data Science and Artificial Intelligence (WSAI) Annual Research Showcase 2025, IIT Madras, Chennai, India. Poster: May 17, 2025.*
4. **Debnath, A.,** Corsi, M-C., Verma, P. (2025). **Biophysical modeling to elucidate the underlying neural mechanisms in EEG-based brain-computer interface.** *Physics of Cells & Tissues (PoCT) 3.0 Symposium, Indian Institute of Science (IISc), Bengaluru, India. Poster: February 21, 2025.*

AWARDS AND HONORS

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| • MS Thesis Research Grant (\$8000 in total) | Summer 2026 |
| • IBSE Post-Baccalaureate Fellowship (\$6000 in total) | 2024–2025 |
| • Paris Brain Institute Travel Award for OHBM 2025 Annual Meeting at Brisbane | 2025 |
| • IBSE Travel Award for Physics of Cells & Tissues 3.0 Symposium at IISc | 2025 |
| • NEC Merit Scholarship to pursue Bachelor's degree (\$1140 in total) | 2020–2024 |

RELEVANT COURSES TAKEN

Graduate Courses:

- **Biomedical Imaging/Bioengineering & Neuroscience:** Principles of MRI – MRI Physics, Systems and Behavioural Neuroscience, Advanced Neurological Imaging, Machine Learning Algorithms for Medical Imaging, Image Processing and Analysis-I & II, Physical Principles of CT, PET and SPECT Imaging, Ultrasound, Imaging Probes for Nuclear and Optical Imaging, Imaging Study Design.

UG Courses:

- **Biotechnology:** Cell Biology and Genetics, Molecular Biology and rDNA Technology, Biochemistry and Enzyme Technology, Microbiology and Bioprocess Technology, Human Genomics, Bioinformatics, Cancer Biology and Cell Signaling, Applications of Molecular Cloning.
- **Mathematics:** Linear Algebra, Calculus, Ordinary Differential Equations, Probability, Partial Differential Equations.
- **Computer Science:** Introduction to Computing, Programming and Data Structures, Compiler Design, Database Management Systems.
- **Others:** Electrical Eng.– Control Systems.